

Rareş Cazan

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Education

Bachelor in Computer Science, POLITEHNICA University of Bucharest

Oct 2024 – Jul 2028 (Expected)

- **Relevant Coursework:** Data Structures and Algorithms, Algorithm Design, Operating Systems, Object-Oriented Programming, Computer Networking, Assembly Language Programming (NASM)

Undergraduate Teaching Assistant, Assembly & Operating Systems

Oct 2025 – Present

- Mentored **15-student** lab sections, providing direct support and evaluating assignments on low-level memory concepts (**NASM**) and Linux system administration (**Bash/CLI**).

Projects

RegAdjudicator

Aug 2026 – Present [🔗](#)

- Engineered an **agentic RAG** system in **Python** with a custom **ReAct orchestration loop**, dynamically routing across FDA/EMA jurisdictions via **Google Gemini** and achieving **~12.5s query latency** using local **ONNX** embeddings.
- Built a two-pass self-verification pipeline with **Pydantic**-structured outputs, auditing claims against **ChromaDB** chunks to trigger reprompt and correct cycles, yielding **100% citation grounding** on a 12-case benchmark.

goMed

Jul 2026 – Present [🔗](#)

- Architected a **Supabase/PostgreSQL** backend and real-time **WebSocket** synchronization, utilizing custom **RPCs** for atomic score updates and achieving **0% dropped packets** at **77ms latency** for 100 concurrent clients.
- Engineered a client-side **CPR pose-tracking** pipeline via **MediaPipe** and **WebAssembly**, computing joint-angle vectors to validate compression depth and 100-120 BPM cadence at **13.5ms/frame** latency.

Ticket Management System

Jan 2026 – Feb 2026 [🔗](#)

- Developed a modular issue-tracking backend in **Java**, leveraging core **Object-Oriented Design Patterns** (Command, Factory, Observer) to safely execute role-based workflows and automate time-sensitive milestones.
- Designed a **Strategy/Visitor**-pattern reporting engine generating 5 distinct report types (customer impact, ticket risk, resolution efficiency, app stability, performance), validated against 19 correctness tests including a 49-step complex scenario, via a custom Jackson **JSON** pipeline.

Async Web Server

Dec 2025 – Jan 2026 [🔗](#)

- Built a concurrent **HTTP/1.1 web server** in **C** using **epoll** multiplexing, **non-blocking sockets**, and **zero-copy** `sendfile()` to maximize static asset throughput.
- Architected a **kernel-level** async I/O (`libaio`) pipeline for dynamic content, integrating `eventfd` and configuring a **128-operation context** to eliminate event-loop blocking.

Volunteering

Embedded Software Engineer at ROSPIN - NOTUS Project

May 2026 – Present

- Co-designed the **core software architecture** and **autonomy stack** for a stratospheric UAV within a **20-person team**.
- Evaluated **FPGA** hardware and core technologies, designing state machines and **RTOS** task scheduling for the implementation phase.

IT Department of Students' League at Computer Science Faculty

Oct 2025 – Present

- Integrated **Google Calendar API** into an internal app, enabling **500+ users** to schedule database-synced meetings with automated **Google Meet** links and **Gmail** alerts.
- Standardized an internal backend project by documenting **REST endpoints** using **Swagger/OpenAPI**.

Certifications & Awards

NXP Cryptographic Algorithms in Embedded Systems

June 2026

- Implemented cryptographic algorithms (**AES-CBC**) in **C** on **NXP LPC55S69** microcontrollers, configuring **USART** peripherals and conducting security tests.

3rd Place, AI Safety & Privacy Research Session

May 2026

- Selected from 30+ projects for proposing a **Zero-Trust privacy architecture** that extracts semantic vectors from text using **NLP**, immediately destroying raw data to match **IEEE P7014** safety guidelines.

Skills and Technologies

- **Languages:** C, Python, Java, C++, Assembly (NASM), Haskell, SQL
- **AI & Machine Learning:** RAG, Vector Databases (ChromaDB), LLM APIs (Gemini), Pydantic, Computer Vision (MediaPipe)
- **Web, Backend & Databases:** React, Node.js, REST APIs, WebSockets, PostgreSQL, Supabase, MongoDB
- **Dev Tools & OS:** Linux, Git, Docker, Postman